

## Lec 1. Basics

US Inflation: Before 1950, average slightly above 0; after 1950, inflation has been the norm.

US unemployment: always exists, no long term trend, correlated to recession

China: recession not necessarily associated with unemployment

Exogenous: taken as given, assumed but not explained

Endogenous: accounted by the model

Market clearing and sticky price:

Market clearing: assumes flexible price, suitable for long-run

Sticky price: suitable for short run

## Lec 2. Data

GDP: Value of all final goods and services

Production = Expenditure = Income

$GDP = C + I + G + EX$

Financing method does not affect GDP

GNP: production by domestic citizens

Capital:  $K_{t+1} = (1 - \delta)K_t + I_t$

Nominal and Real GDP: Current/base price

GDP deflator = Nominal / Real  $\times 100$

$CPI = 100 \times \text{Cost of basket in current price} / \text{Cost of basket base price}$

CPI Bias: Substitution effect, new goods, quality changes

CPI vs GDP Deflator:

Capital goods excluded in CPI

Imported goods included in CPI

Basket fixed in CPI

Employment:

Labor force: employed + unemployed

Unemployed: not employed + actively seeking work

Unemployment Rate =  $100 \times \text{Unemployed} / \text{Labor Force}$

Labor force participation =  $100 \times \text{Labor Force} / \text{Total Population}$

## Lec 3. National Account

Production:  $Y = F(K, L)$

Constant returns to scale

Factor payments:

Wage = MPL, Rental Rate = MPK

Profit =  $Y - WL - RK$

Under constant returns:  $Y = WL + RK$

Cobb-Douglas Y: Labor/capital share constant

Consumption:  $C = C(Y - T)$

Investment:  $I = I(r)$ ,  $r$  is real interest rate

Equilibrium:

Goods market:  $C(Y - T) + I(r) + G = Y$

Loanable funds:  $I(r) = Y - C(Y - T) - G$

## Lec 4. Growth

Solow-Swan model:

Without population growth:  $\Delta k = sf(k) - \delta k$

Unconditional convergence: steady state  $k$ , convergence at decreasing speed

Conditional convergence: controlling for same  $k$ , poorer countries grow faster

Golden rule: maximize consumption  $c = f(k^*) - \delta k^*$  where  $MPK = \delta$

No automatic convergence, requires government adjustment of  $s$

With population growth:  $\Delta k = sf(k) - (\delta + n)k$

With technical improvement ( $E$  grows at  $g$ ):  $\Delta k = sf(k) - (\delta + n + g)k$

Properties:  $K/Y$  constant, real wage grows with  $Y/L$ , real rental price constant

GDP growth decomposition:

$$\frac{\Delta Y}{Y} = \frac{\Delta A}{A} + \alpha \frac{\Delta K}{K} + (1 - \alpha) \frac{\Delta L}{L}$$

$$\frac{\Delta(Y/L)}{Y/L} = \frac{\Delta A}{A} + \alpha \frac{\Delta(K/L)}{K/L}$$

$$\text{Hall and Jones: } \frac{Y}{L} = \left(\frac{K}{Y}\right)^{1-\alpha} hA^{\frac{1}{1-\alpha}}$$

AK model:  $Y = AK$ , permanent growth if  $sA > \delta$  at rate  $sA - \delta$

## Lec 5. Money

Money supply:

Total money:  $M = C + D$

Monetary base:  $B = C + R$

$cr = C/D$ ,  $rr = R/D$

$$M = \frac{cr + 1}{cr + rr} \times B$$

Capital adequacy ratio: bank's equity / total assets

Fed policies:

Open market operations (buy/sell government bonds)

Discount rate adjustment

Reserve requirements

Interest on reserves

Quantitative Easing (buy long-term bonds)

Quantity theory:  $MV = PY$

Velocity  $V$  exogenous

$$\text{Inflation: } \pi = \frac{\Delta M}{M} - \frac{\Delta Y}{Y}$$

Fisher effect:  $i = r + \mathbb{E}\pi$

Money demand:  $M/P = kY$  or  $M/P = L(i, Y)$

$L$  decreases with  $i$

Classical dichotomy: nominal variables don't affect real variables

## Lec 6. Unemployment

Hours worked per capita = Hours per worker  $\times$  (1 - unemployment rate)  $\times$  participation rate

Surveys:

Current Population Survey (household)

Establishment Survey (firms)

Unemployment components: cyclical, frictional, structural

Structural unemployment: demand and supply mismatch

Unemployment dynamics:

$$\Delta U = bN + s(N - U) - (d + f)U$$

$$\Delta N/N = b - d$$

Steady state:  $u^* = \frac{b + s}{b + s + f}$  (frictional unemployment)

Job finding rate:  $f = f(v, i, m)$

$v$ : vacancies,  $i$ : search intensity,  $m$ : matching ease

Beveridge curve:  $u$ - $v$  relationship

Job-creation curve:  $v = v(u)$  with  $v' > 0$

Higher  $u$  leads to lower wages and more vacancies

## Lec 8. Aggregate Supply

### Lec 7. Short-run Aggregate Demand: IS-LM Curve

Business cycle: Expansion-Contraction

IS curve:  $Y = C + I + G$

$C = a + \bar{b}(Y - T)$

$I = c - \bar{d}r$

$G = \bar{G}, T = \bar{T}$

Multipliers:  $\Delta Y = \frac{1}{1-b} \Delta G = -\frac{b}{1-b} \Delta T$

IS equation:  $r = -\frac{1-\bar{b}}{\bar{d}}Y + \frac{a+c+G-\bar{b}T}{\bar{d}}$

LM curve:  $MV = PY, V = V(i)$

$M = \bar{M}, i = r + \mathbb{E}\pi$

$L(r + \mathbb{E}\pi, Y) = Y - \bar{k}(r + \mathbb{E}\pi)$

LM equation:  $r = \frac{1}{\bar{k}}Y - \mathbb{E}\pi - \frac{M}{\bar{k}\bar{P}}$

AD curve: Derived from IS-LM, downward sloping in  $P$ - $Y$  space

Long run:  $Y$  constant

Short run:  $P$  constant

Short-run AS:  $Y = \bar{Y} + \alpha(P - \mathbb{E}P)$

Derivation:  $s$  firms have sticky prices

Others set  $p^* = P + \bar{z}(Y - \bar{Y})$

$\alpha = \frac{s}{(1-s)\bar{z}}$

Higher inflation makes AS steeper

Phillips curve:  $\pi = \mathbb{E}\pi - \beta(u - u^n) + v$

Sacrifice ratio: unemployment needed to reduce inflation

Taylor's rule:  $i_t = 2\% + \pi_t + 0.5(\pi_t - 2\%) + 0.5(Y_t - \bar{Y}_t)$

## Lec 9. Fiscal Policy and Consumption

Utility:  $U(c_1, c_2) = \ln(c_1) + \ln(c_2)$

Budget constraint:  $c_1 + s = y_1$

$c_2 = y_2 + (1+r)s$

With taxes: disposable income  $y_i - \tau_i$